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ION PRODUCTION SYSTEM WITH FIBROUS LATTICE FOR ION COLLECTION

Abstract

A method that includes accelerating ions toward a lattice of carbon fibers and capturing the ions in the lattice of carbon fibers.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/819,854, filed Aug. 29, 2024, which is a continuation of U.S. patent application Ser. No. 17/957,310, filed Sep. 30, 2022, issued as U.S. Pat. No. 12,112,859, which claims the benefit of U.S. Provisional Application No. 63/251,397 filed on Oct. 1, 2021, which is incorporated herein by reference in its entirety.

TECHNOLOGY

[0002] The present disclosure relates generally to the field of heavy metal ion production, for example production of heavy metal ions used in health care applications. More specifically, the present disclosure relates to improving the efficiency of collecting and constituting ions accelerated out of an ion source at a high extraction energy.

BACKGROUND

[0003] Previous technologies that involve ion beams are typically intended to provide high-energy collisions between the ion beam and a substrate material, in order to create changes to the substrate material. The ions themselves in such systems are not efficiently retained in the substrate material, and they may be sputtered, sublimated, or scattered away. Such systems and methods do not provide for high-efficiency collection of the ions themselves. In contrast, one goal of the present application is collection of the ions of the ion beam as a constituted material which can be collected, stored, transported, used, etc., for example in health care applications.

SUMMARY

[0004] One implementation of the present disclosure is a method. The method includes providing a lattice of carbon fibers, accelerating ions toward the lattice of carbon fibers, and capturing the ions in the lattice of carbon fibers. The method may also include burning the lattice of carbon fibers to obtain a residue comprising the ions.

[0005] The ions may be ytterbium ions. In some embodiments, the ions include ytterbium-176 ions and the method also includes isolating the ytterbium-176 ions from other isotopes before capturing the ions in the lattice of carbon fibers.

[0006] In some embodiments, accelerating the ions toward the lattice of carbon fibers includes providing the ions with energies greater than 100V. Capturing the ions in the lattice of carbon fibers may include decelerating the ions by deflecting the ions off of a plurality of the carbon fibers of lattice of carbon fibers.

[0007] In some embodiments, the method includes increasing an area of the lattice of carbon fibers that captures the ions by operating an actuator to rotate or translate the lattice of carbon fibers. The method may also include providing multiple layers of a fibrous carbon material. Providing the lattice of carbon fibers may include arranging the carbon fibers in a plurality of directions.

[0008] Another implementation of the present disclosure is an ion production system. The ion production system may include an ion source configured to produce ions, a target comprising a fibrous lattice, and an electrode positioned between the ion source and the target substrate and configured to accelerate the ions toward the target substrate so that the ions are incident on the fibrous lattice. The fibrous lattice is configured to capture the ions. The ions may be ytterbium ions, for example including ytterbium-176 ions.

[0009] The fibrous lattice may include a plurality of carbon fibers arranged in a plurality of directions. The target may include a plurality of layers of the fibrous lattice. The fibrous lattice may include graphite or carbon, and may be configured to burn. The fibrous lattice may leave a residue comprising the ions after burning of the fibrous lattice.

[0010] In some embodiments, the electrode provides the ions with energies greater than 100V. In some embodiments, the target includes a mount configured to releasably secure the fibrous lattice in position relative to the mount. The ion production system may include an actuator operable to rotate the target.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0011] The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:

[0012] FIG. 1 is a schematic diagram of an ion production system, according to an illustrative embodiment.

[0013] FIG. 2 is a schematic diagram of a target and voltage source of the ion production system in an embodiment involving positive ions, according to an illustrative embodiment.

[0014] FIG. 3 is a schematic diagram of a target and voltage source of the ion production system in an embodiment involving negative ions, according to an illustrative embodiment.

[0015] FIG. 4 is a block diagram of an ion production system having a magnetic rotation device, according to some embodiments.

[0016] FIG. 5 is an exploded view of a fibrous lattice of a target of an ion production system, according to some embodiments.

[0017] FIG. 6 is a perspective view of a fibrous lattice of a target of an ion production system, according to some embodiments.

[0018] FIG. 7 is a top view of target of an ion production system, according to some embodiments.

[0019] FIG. 8 is a top view of a fibrous lattice of a target of an ion production system,

[0020] according to some experimental results.

[0021] FIG. 9 is a top view of a fibrous lattice of a target of an ion production system, according to some experimental results.

DETAILED DESCRIPTION

[0022] Before turning to the figures, which illustrate certain embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0023] Referring generally to the figures, systems and methods relating to an ion production system, for example a heavy metal ion production system, are shown, according to various example embodiments. In particular, the figures generally show systems and methods relating to high-efficiency collection of ions (e.g., heavy metal ions such as ytterbium ions including ytterbium-176 ions) at a target of the ion production system, such that the ions are reconstituted as a material which can then be collected, stored, transported, used, etc. for various applications (e.g., a material having a high concentration of ytterbium-176 or other desired isotope).

[0024] As detailed below, the ions are generated in an ion source and extracted out of the ion source as an ion beam having a high extraction energy, for example between 20 kV and 80 kV (e.g., between 40 kV and 60 kV) (noting that ion kinetic energy is often expressed in volts in this context, with one volt equaling one joule per coulomb; in other conventions these values may be expressed in keV). The ion beam may be passed through a magnetic analyzer, which uses a magnetic field to sort ions by momentum (or, if the charges of all the generated ions are the same, by atomic mass), and a mass-resolving aperture which is arranged relative to the beam and the magnetic analyzer to primarily pass desired ions (e.g., ions of a desired isotope) while blocking other ions from passing through the aperture. An ion beam having only the desired ions or a high percentage of the desired ions is thus passed through the aperture. Thus, the mass-resolving aperture isolates the desired ions or a high percentage of the desired ions. The target (target substrate, substrate, etc.) is positioned such that the beam of desired ions is incident on the target.

[0025] The approaches described herein provide for a high rate of ion sticking to the target and low sputtering rates at the target, thus increasing the percentage of the ions that are constituted as a

neutral material (e.g., of the desired isotope) and decreasing the percentage of the ions which are lost. The overall efficiency (e.g., power and resource consumption per unit of material produced) of the whole ion production system are thus improved. Build-up of the ions on other, undesired surfaces of the ion production system (which may otherwise result from scattering of ions) is also minimized or prevented, thereby reducing downtime and maintenance of the ion production system. Additionally, the approaches herein can reduce heat transfer to the target substrate (heat load on the target, thermal energy load on the target) and thus reduce or eliminate a temperature management issue that may otherwise be caused by high-energy collisions at the target.

[0026] As detailed below, some of the advantages herein are achieved in part and/or in some embodiments, by holding the target at a voltage having the same polarity as the ions and having a magnitude slightly below the potential of the ion beam. The voltage of the target (and the electric field created by that voltage) causes a reduction in energy of the ion beam as the ion beam approaches the target. For example, the voltage of the target may be selected to offset (at least partially) the energy added to the ion beam provided by the extraction electrodes, such that the energy of the ions is reduced to a thermal energy as the ions reach the target. In such cases, both electronic stopping (i.e., interactions between the ions and the target interactions between electrons of the ions and electrons of the target substrate) and nuclear stopping (i.e., interactions between the nuclei of the ions and nuclei of the target substrate) of the ions at the target are reduced to zero, or near-zero, levels. By reducing these interactions via the potential of the target, the ions are caused to stick and form as a film at the target rather than colliding with the target at high energies and sputtering or scattering away.

[0027] The systems and methods described herein also provide a material of the target which is well-adapted to efficiently capture ions incident thereon and to be easily reduced to a residue having a high concentration of a desired isotope/atom (i.e., from the ion beam). In particular, as described in detail below, a fibrous lattice, for example a lattice, felt, mesh, etc. of carbon fibers (e.g., graphite fibers) can be positioned so that the ion beam is incident thereon. The fibrous lattice captures the ions and can be easily burned (or subjected to some other reaction) when in the presence of oxygen to leave a residue having a high concentration of a desired isotope, for example. The fibrous lattice can be rotated or translated relative to the beam during operation of the ion production system to increase the collection capacity of the target before operation needs to be paused to harvest the desired isotope from the target. The various concepts described below thus individually and collectively provide for highly efficient collection of a desired isotope.

[0028] Referring now to FIG. 1, a block diagram of an ion production system **100** is shown, according to an illustrative embodiment. The ion production system **100** includes an ion source **102**, extraction electrode(s) **104**, a magnetic analyzer **106**, a mass-resolving aperture **108**, a target **110**, and a voltage source **112** connected to ground **114** and to the target **110**. The ion source **102**, the extraction electrode(s) **104**, the magnetic analyzer **106**, the mass-resolving aperture **108**, and the target **110** are arranged sequentially such that ions are generated at the ion source **102** and sequentially pass the extraction electrode(s) **104**, magnetic analyzer **106**, and mass-resolving aperture **108** before reaching the target **110**. As detailed below, the ion production system **100** is configured to provide efficient collection of desired ions at the target **110** as a constituted, neutral material which can be removed, stored, transported, etc., and eventually used for some application, for example a healthcare application.

[0029] The ion source **102** is configured to produce ions. For example, the ion source **102** may be configured as a Bernas or Freeman ion source, which includes a filament operable to emit electrons which ionize a gas provided into the ion source **102**, for example a heavy metal gas such as a ytterbium vapor. Other metals (Lu, Tc, etc. may also be used). Interactions between the electrons and the gas ionize the gas to produce ions. In some embodiments, the ion source **102** produces positive ions (i.e., “cations,” ions having a positive polarity). In other embodiments, the ion source produces negative ions (i.e., “anions,” ions having a negative polarity). The ion source **102** includes

an outlet slit or aperture so that the ions can be extracted from the ion source **102**. In some embodiments, the ion source **102** includes auxiliary heaters to protect elements of the ion source **102** and to improve uniformity of the ions for extraction from the ion source **102**, for example as described in detail in U.S. Provisional Patent Application No. 63/122,699, filed Dec. 8, 2020, the entire disclosure of which is incorporated by reference herein.

[0030] The extraction electrode(s) **104** includes one or more electrodes that are configured and operated to provide an electric field that extracts the ions from the ion source **102**. Because the ions have an electric charge of a first polarity (positive or negative in different embodiments), a voltage of the opposite polarity at the extraction electrode(s) **104** will pull the ions out of the ion source as an ion beam. The extraction electrode(s) **104** may include one or more electrodes to accelerate the ion beam, decelerate the ion beam, shape the beam, aim the beam, etc. By providing an electric field that accelerates the ion beam out of the ion source **102**, the extraction electrode(s) **104** provide the ion beam with an extraction energy of the same or similar magnitude as a voltage of the extraction electrode(s) **104**. For example, an electrode at a voltage of 55 kV can provide the ion beam with an extraction energy of 55 kV (noting that ion kinetic energy is often expressed in volts in this context, with one volt equaling one joule per coulomb) as the ion beam pass the extraction electrode(s) **104**.

[0031] An ion beam having a high extraction energy is thus provided as an output of the extraction electrode(s) **104**. In various embodiments, the high extraction energy can be in a range between 20 kV and 80 kV, for example between 40 kV and 60 kV (e.g., 55 kV). In such embodiments, the voltage applied at the extraction electrode(s) **104** can be selected to provide the ion beam with the desired extraction energy for a particular scenario.

[0032] In the example of FIG. 1, the ion beam passes from the extraction electrodes **104** to the magnetic analyzer **106**. In other embodiments, the magnetic analyzer **106** is omitted. The magnetic analyzer **106** is configured to provide a magnetic field that creates magnetic forces on the ion beam. The magnetic force on each ion may be approximately equal, but the ion beam may include ions of different isotopes, such that the masses of the ions vary. The magnetic force provided by the magnetic analyzer **106** may result in a separation of ions by mass. Thus, after passing through the magnetic analyzer **106**, different areas of a transverse cross-section of the ion beam may include different isotopes, i.e., ions of different mass.

[0033] In FIG. 1, the ion beam is illustrated to pass from the magnetic analyzer **106** to the mass-resolving aperture **108**, and the mass-resolving aperture **108** is configured to block an undesired subset of the ions from passing through the mass-resolving aperture **108**, while allowing desired ions to pass through the mass-resolving aperture **108**. Thus, the mass-resolving aperture **108** isolates the desired ions or a high percentage of the desired ions. In particular, ions allowed through the mass-resolving aperture **108** are primarily ions of a desired isotope (or two desired isotopes), while ions of one or more other isotopes are intercepted by the mass-resolving aperture **108**. This is achieved by positioning the mass-resolving aperture **108** relative to the magnetic analyzer **106** to take advantage of the separation of isotopes by mass achieved by the magnetic analyzer **106**. Various geometric arrangements are possible in various embodiments. Thus, in examples including the magnetic analyzer **106** and the mass-resolving aperture **108**, the ion beam that reaches the target **110** includes a high percentage of a desired isotope or desired isotopes, with a low percentage of contamination by ions of different isotopes.

[0034] The ion beam from the mass-resolving aperture **108** is incident on target **110**. The target **110** is configured to receive and collect the ions of the ion beam. The target **110** can include a substrate material suitable for receiving and retaining the ions, including as a film on a surface of the target **110** and/or embedded in a lattice structure of the target **110**. For example, the substrate material of the target **110** may have a crystal structure. As another example, the substrate material of the target **110** may include a carbon fiber material (e.g., a carbon fiber cloth). The material(s) of the target **110** are also selected such that the target **110** is capable of being held at a substantially-constant

voltage as ions are collected on, implanted in, or other received at the target **110**. The material(s) of the target **110** may be selected to help cause sticking of the ions to or in the target **110**. The target **110** may be removable and replaceable in the ion production system **100** to facilitate harvesting of the ionic material that builds up on the target **110** during operation of the ion source **102**.

[0035] The target **110** is shown as being coupled to a voltage source **112**, which is connected between the target **110** and ground **114**. Other elements of the ion production system **100** also include suitable electronics elements, power sources, etc. to enable operation thereof. The voltage source **112** is configured to hold (put, establish, maintain, etc.) the target **110** at a voltage (referred to herein as the target voltage) which has the same polarity of the ion beam. For example, as illustrated in FIGS. 2-3 and discussed below, a positive voltage is provided to the target **110** by the voltage source **112** in scenarios where positive ions are generated by the ion source, while a negative voltage is provided to the target **110** by the voltage source **112** in scenarios where negative ions are generated by the ion source. In other embodiments, the target **110** is connected directly to ground such that the target **110** is not biased as in the examples of FIGS. 2-3.

[0036] The target voltage is preferably less than the extraction energy of the ion beam, such that the ion beam is able to reach the target without being forced in the opposite direction by the target voltage, while being high enough to reduce the energy of ion beam far enough to minimize both electronic and nuclear stopping of the ion beam at the target **110** (thereby minimizing scattering or sputtering that would otherwise be caused by high-energy collisions between the ions and the target **110**). For example, the target voltage may be less than the extraction energy by an amount corresponding to a thermal energy of the ions, such that the energy of the ions is reduced to thermal energy just as the ions reach the target **110**. In various embodiments, the target voltage is both less than the extraction energy and greater than 95% of the extraction energy, for example greater than 99% of the extraction energy (while also being less than the extraction energy). In some examples, the target voltage is approximately 100 V less than the extraction energy such that the ion beam has an energy of approximately 100 V when the ion beam reaches the target (e.g., extraction energy minus target voltage equals approximately 100V). In one example, the extraction energy is 55 kV and the target voltage is 54.9 kV.

[0037] In some embodiments, the voltage source **112** and the target **110** are configured so that the voltage of the target **110** stays substantially constant throughout operation of the ion production system **100** and as ions collect on the target **110** (e.g., as a film on the target **110**, embedded in the target **110**) and are constituted into a neutral material (e.g., of the desired isotope(s)). In some cases, to facilitate removal of the constituted ionized material from the target **110**, the target **110** can be removable from the ion production system **100**. In some such cases, the voltage source **112** is controlled to gradually reduce the target voltage toward zero to enable the target **110** to be disconnected from the voltage source **112** without disrupting the ionized material collected thereon. In some embodiments, the target **110** (or a portion thereof) is removed for use in transport and further processing of the ionized material, and replaced with a new target **110** (or new portion thereof) for use in subsequent operation of the ion production system **100**. In other embodiments, the ionized material can be removed from the target **110** and collected in a receptacle (or other collection and retention device) such that the target **110** can be reused in a subsequent operation of the ion production system **100** to collect more ions.

[0038] Referring now to FIG. 2, a schematic illustration of the target **110** and voltage source **112** of the ion production system **100** in an embodiment involving positive ion beam **150** is shown, according to an example embodiment. FIG. 2 shows a positive ion beam **150** (i.e., a beam of positively-charged ions) aimed towards the target **110** and incident on the target **110**.

[0039] Because the positive ion beam **150** has a positive polarity, the voltage of the target **110** is also provided with a positive polarity. FIG. 2 illustrates that the target **110** is connected to a positive terminal of the voltage source **112**, with the negative terminal of the voltage source **112** connected to ground **114**. The voltage source **112** holds the target **110** at a positive potential, i.e., an electrical

potential of the same polarity as the positive ion beam **150**.

[0040] The positive potential of the target **110** provides an electric field that resists movement of the ion beam **150** toward the target **110**. The ion beam **150** must move across this electric field to reach the target **110**. As it does so, kinetic energy of the ion beam **150** is converted into electric potential of the ions in the electric field created by the positive potential of the target **110**. This may be thought of as being analogous to the ions rolling “uphill” to reach the target **110**. As discussed above, the target voltage is selected and maintained such that the positive ion beam **150** reaches a low energy, for example a thermal energy, just as positive ions reach the target **110**. Reduced to thermal energy, the positive ion beam **150** does not have extra kinetic energy that would cause it to move away from the target **110** or cause sputtering or scattering, and, thus, the ions of the positive ion beam **150** stick at the target **110**, for example forming as the positive ion film **151** as shown in FIG. 2.

[0041] Referring now to FIG. 3, a schematic illustration of the target **110** and voltage source **112** of the ion production system **100** in an embodiment involving negative ion beam **152** is shown, according to an example embodiment. FIG. 2 shows a negative ion beam **152** (i.e., a beam of negatively-charged ions) aimed towards the target **110** and incident on the target **110**.

[0042] Because the negative ion beam **152** has a negative polarity, the voltage of the target **110** is also provided with a negative polarity. FIG. 3 illustrates that the target **110** is connected to a negative terminal of the voltage source **112**, with the positive terminal of the voltage source **112** connected to ground **114**. The voltage source **112** holds the target **110** at a negative potential, i.e., an electrical potential of the same polarity as the negative ion beam **152**.

[0043] The negative potential of the target **110** provides an electric field which resists movement of the ion beam **152** toward the target **110**. The ion beam **152** must move across this electric field to reach the target **110**. As it does so, kinetic energy of the ion beam **152** is converted into electric potential of the ions in the electric field created by the negative potential of the target **110**. This may be thought of as being analogous to the ions rolling “uphill” to reach the target **110**. As discussed above, the target voltage is selected and maintained such that the negative ion beam **152** reaches a low energy, for example a thermal energy, just as negative ions reach the target **110**. Reduced to thermal energy, the negative ion beam **152** does not have extra kinetic energy that would cause it to move away from the target **110** or cause sputtering or scattering, and, thus, the ions of the negative ion beam **152** stick at the target **110**, for example forming as the negative ion film **153** shown in FIG. 3.

[0044] The ion production system **100** is thereby configured for highly efficient production and collection of ions as a constituted ionized material. By setting the target **110** at the target voltage as described above, a high percentage of the ions incident on the target **110** are caused to stick at the target **110**, for example forming a film at the target **110**. The efficiency of the ion production system **100** is thereby improved by providing collection of a high percentage of the desirable ions created by the ion source **102**. Additionally, because material is sputtered or scattered away at a low or zero rate, it is also substantially prevented from building up on other, unwanted surfaces in the ion production system **100**, thereby reducing downtime, cleaning, maintenance, etc. of the ion production system **100**. Additionally, while electronic or nuclear stopping of high energy ions at the target (i.e., collisions between atoms) would cause the thermal energy of the target to increase greatly, the embodiments herein reduce the energy of the ions using the electric potential provided by the voltage source **112** and thereby avoid the build-up of thermal energy at the target.

[0045] Referring now to FIG. 4, a schematic diagram of an apparatus **200** that includes a vacuum chamber **202**, an ion production system **204**, and magnetic rotation device **206** is shown, according to some embodiments.

[0046] The ion production system **204** is shown as including an ion beam generator **208** and a target **210**. The ion beam generator **208** is configured to generate an ion beam and direct the ion beam toward the target **210** so that the ion beam is incident on the target **210**. The ion beam

generator **208** can include the ion source **102**, the extraction electrodes **104**, the magnetic analyzer **106**, and/or the mass-resolving aperture **108** of FIG. **1**, for example. The target **210** may be configured the same as the target **110** of FIG. **1**, for example.

[0047] As shown in FIG. **4**, the target **210** is coupled to the magnetic rotation device **206**. The magnetic rotation device **206** includes an interior plate (first plate) **212**, and exterior plate (second plate) **214**, and a motor **216**. The interior plate **212** is inside (interior to, internal to, contained in) the vacuum chamber **202** and is positioned on an interior side **218** of a wall **220** of the vacuum chamber **202**. The exterior plate **214** is outside (exterior to, external to, not contained in) the vacuum chamber **202** and is positioned on an exterior side **222** of the wall **220**. The motor **216** is mechanically coupled to the exterior plate **214**. The target **210** is mechanically coupled to the interior plate **212**.

[0048] The motor **216** is operable to drive rotation of the exterior plate **214**. The motor **216** may be an electric motor, for example a stepper motor, which transforms electricity into rotational movement. The motor **216** is coupled to the exterior plate **214** such that operation of the motor **216** exerts a torque on the exterior plate **214** that causes rotation of the exterior plate **214** about an axis of the exterior plate **214**. A rotating drive shaft of the motor **216** can be aligned with the axis of the exterior plate **214** to directly transfer the torque to the exterior plate **214** to cause rotation of the exterior plate **214**. The motor **216** can be controllable to rotate the exterior plate **214** at various speeds. In some embodiments, the motor **216** is operated to rotate the exterior plate **214** at a pace of approximately one rotation per minute.

[0049] The exterior plate **214** includes one or more magnets (e.g., permanent magnets) and the interior plate **212** also includes one or more magnets (e.g., permanent magnets) corresponding to the one or more magnets of the exterior plate **214**. An example arrangement of magnets in the interior plate **212** and the exterior plate **214** is shown in FIGS. **3-6** and described with reference thereto. The magnets of the exterior plate **214** and the interior plate **212** are arranged such that an attractive force is exerted on the interior plate **212** by the exterior plate **214** and vice versa. For example, the one or more magnets of the exterior plate **214** may be arranged with a positive magnetic polarity facing the wall **220** while the one or more magnets of the interior plate **212** are arranged with a negative magnetic polarity facing the wall **220** (or vice versa), such that the magnets are attracted toward one another and a magnetic force pulls the interior plate **212** and the exterior plate **214** together. The magnets provide sufficiently strong magnetic fields to exert the attractive force across the wall **220** of the vacuum chamber **202**. The wall **220** may be approximately half of an inch thick in some embodiments.

[0050] Due to the attractive magnetic force between the magnets of the exterior plate **214** and the interior plate **212**, rotation of the exterior plate **214** by the motor **216** causes rotation of the interior plate **212**. In the examples shown, the interior plate **212** rotates to match the rotation of the exterior plate **214** due to magnetic coupling therebetween. Rotational movement and torque (e.g., angular kinetic energy) is thereby communicated across the wall **220** of the vacuum chamber **202** without compromising the integrity of a hermetic seal of the vacuum chamber **202** (e.g., without requiring mechanical engagement between the interior plate **212** and the exterior plate **214** that may be difficult to hermetically seal). Because, as shown in FIG. **4**, the target **210** is mounted on the interior plate **212**, rotation of the interior plate **212** rotates the target **210**. Although the examples herein refer to rotation, in other embodiments the motor **216** is arranged to translate the exterior plate **214** (e.g., in one or two dimensions) to thereby cause corresponding translation of the interior plate **212** and the target **210**. Operation of the motor **216** thereby causes motion of the target **210**, for example rotation of the target **210**.

[0051] As illustrated in FIG. **4**, the ion beam generator **208** directs the ion beam at the target **210** so that the ion beam is misaligned (offset, etc.) relative to an axis of rotation of the target **210**. Accordingly, when the target **210** rotates by operation of the magnetic rotation device **206**, the point or area at which the ion beam is incident on the target **210** changes. Rotation of the target **210**

over time causes the ion beam to be incident on different portions of the target **210** over time, thereby increasing a total area of the target **210** that is exposed to the ion beam. Movement of the target **210** thereby allows a larger area of the target **210** to be exposed to the ion beam and to capture ions and/or isotopes from the ion beam. The target **210** can thus capture more material as compared to an embodiment with a static target **210**, allowing for longer continuous operation of the apparatus **200** before the target is full (saturated, at capacity, etc.). Rotating the target can also help reduce temperature gradients across the target, which may be undesirable.

[0052] The magnetic rotation device **206** is also configured to provide heat transfer into or out of the vacuum chamber, for example to remove heat from the target **210** to manage the temperature thereof. As shown, the interior plate **212** and the exterior plate **214** are both positioned in contact with the wall **220** of the vacuum chamber **202**. The interior plate **212** and the exterior plate **214** can include a material with high thermal conductivity (e.g., low resistance to heat flow therethrough), for example a metal such as steel. The wall **220** may be made of a similar material. The interior plate **212** and the exterior plate **214** are in thermal contact with one another via the wall **220**. Such thermal contact is maintained by the attractive force between the magnets of the interior plate **212** and the exterior plate **214** which can force the interior plate **212** and the exterior plate **214** towards one another and into contact with the wall **220**. The target **210** shown as positioned on the interior plate **212**. A pathway for heat transfer is thereby provided from the target **210** to the exterior plate **214**.

[0053] In the embodiment shown, the apparatus **200** also includes a cooling system **224** in thermal communication with the exterior plate **214**. The cooling system **224** can include a refrigeration cycle (including a compressor, condenser, expansion valve, and evaporator, for example) configured to remove heat from the exterior plate **214**. For example, the cooling system **224** may provide a chilled fluid through one or more coils or other heat exchanger in thermal contact with the exterior plate **214**. Cooling of the exterior plate **214** increases the heat flow away from the target **210**, which may be desirable in embodiments where collision of the ion beam with the target **210** provides thermal energy to the target **210**. In other scenarios (e.g., other use cases for the magnetic rotation device **206**), the cooling system **224** can include or be replaced with a heating system configured to provide thermal energy to the exterior plate **214** in order to transfer thermal energy (heat) into the vacuum chamber **202** via the interior plate **212**.

[0054] Referring now to FIG. 5, an exploded view of the target **110** (or target **210**) or a portion thereof (e.g., a fibrous lattice thereof) is shown, according to some embodiments. In the example of FIG. 5, the target **110** includes a first lattice **400** and a second lattice **402** which form the target **110** as a fibrous lattice. The first lattice **400** and the second lattice **402** may be stacked as layers to form the target **110**. In other embodiments, other numbers of lattices (layers) are included in the target **110** (e.g., one, three, four, five, etc.). The fibrous lattice may be formed as a carbon felt or carbon foam in various embodiments.

[0055] The first lattice **400** includes a plurality of fibers arranged in a plurality of directions, shown as two orthogonal directions. The plurality of fibers may be woven together or otherwise coupled to form the first lattice **400**. The second lattice **402** also includes a plurality of fibers arranged in a plurality of directions, shown as two orthogonal directions, which are woven together or otherwise coupled to form the second lattice **402**. The first lattice **400** and the second lattice **402** may be arranged relative to one another such that fibers of the first lattice **400** are parallel with fibers of the second lattice **402**, or may be oriented differently so that fibers of the first lattice **400** are at non-orthogonal angles relative to fibers of the second lattice **402**. In some embodiments, the first lattice **400** and the second lattice **402** appear substantially solid to the naked human eye, but are made up of fibers at a microscopic or smaller level.

[0056] The fibers of the first lattice **400** and the second lattice **402** may be made of carbon, for example such FIG. 5 shows a first lattice **400** of carbon fibers and a second lattice **402** of carbon fibers. In some embodiments, the fibers are made of graphite, for example such that some or all of

the fibers of the first lattice **400** and the second lattice **402** are graphite fibers. The material of the fibers is preferably a high purity (e.g., greater than 95% carbon) such that, when burned (in the presence of oxygen), the carbon fibers themselves leave little or no solid residue. When in vacuum (e.g., when inside the vacuum chamber **202** during operation of the apparatus **200**, substantially outside the presence of oxygen) the carbon fibers are configured to handle high temperatures (e.g., greater than 200 C, greater than 300 C, greater than 800 C) without substantially deforming, melting, etc.

[0057] The first lattice **400** and the second lattice **402** are configured to capture ions incident thereon (e.g., ions from the ion beam provided by the ion beam generator **208** of FIG. **4**). The arrangement of the plurality of fibers causes an ion to deflect (scatter, collide, etc.) off of multiple fibers as the kinetic energy of the ion is reduced until the ion will stay at the target **110** (e.g., reduce to thermal energy), without scattering away from the target **110** after a single collision. The arrangement of the fibers is partially porous, such that some ions are able to penetrate beyond an outer surface of the first lattice **400**, thus reducing the amount of energy built up at the surface of the first lattice **400** and allowing ions to scatter multiple times without escaping the target **110** (e.g., vaporizing away from the target **110**). Accordingly, relative to a flat plate or block of material, the lattice structure provides an increased surface area and overlapping geometry that can facilitate capture a high percentage of the ions incident on the target **110** (e.g., greater than 40%, greater than 90% in some arrangements). The first lattice **400** and the second lattice **402** thereby provide efficient collection of the desired isotope at the target **110**.

[0058] The first lattice **400** and the second lattice **402** are also configured to burn (in the presence of oxygen) or otherwise react to leave (e.g., be reduced to) a residue that includes a high concentration of the desired isotope. For example, the fibrous lattice configuration of FIG. **5** provides the target **110** with an increased surface-area-to-mass ratio as compared to a solid block or plate of carbon or graphite, which enables relatively easy burning of the of the first lattice **400** and the second lattice **402** (e.g., as compared to a solid block of graphite, which typically will not burn). For example, during operation of the ion production system **100**, the target **110** captures the desired isotope (in the first lattice **400** and the second lattice **402** in the example of FIG. **5**) while the target **110** is held in vacuum without a substantial amount oxygen present (thereby preventing complete burning of the target **110**). The target **110** can then be removed from vacuum (e.g., target **210** can be removed from the vacuum chamber **202** in the example of FIG. **4**) for processing to extract the isotope from fibrous lattice. Outside vacuum, oxygen is present which allows burning of the carbon fibers. The fibrous lattice (e.g., the first lattice **400** and the second lattice **402**) can then be burned to reduce the fibrous lattice to a residue having a high concentration of the desired isotope. The carbon dissipates as gas after burning, such that the remaining material is of the desired isotope, which may oxidize during the extraction process. For example, in some embodiments, a powder of oxidized ytterbium (e.g., oxidized ytterbium-176) is left as a powder (e.g., white-colored powder) after burning of the target.

[0059] Referring now to FIG. **6**, a perspective view of a fibrous lattice **500** of the target **110** (or target **210**) is shown, according to some embodiments. The fibrous lattice **500** can be used as an alternative to the first lattice **400** and the second lattice **402** of FIG. **5**, or may be used in combination with the first lattice **400** and/or the second lattice **402** of FIG. **5** in various embodiments.

[0060] As shown in FIG. **6**, the fibrous lattice **500** includes a plurality of fibers arranged in a tangled web, such that the fibrous lattice **500** may be characterized as an open-celled foam. As in the example of FIG. **5**, the plurality of fibers may be carbon fibers and/or graphite fibers. The fibrous lattice **500** is configured to capture ions (e.g., from the ion beam generated by the ion beam generator **208** in the example of FIG. **2**) such that the desired isotope is collected in the fibrous lattice **500**. The fibrous structure of the fibrous lattice **500** may cause ions to deflect off multiple fibers before coming to rest in the fibrous lattice **500**, without scattering away from the fibrous

lattice **500** after a single collision. The fibrous lattice **500** also has a high surface-area-to-mass ratio which facilitates easy burning of the fibrous lattice **500** to reduce the fibrous lattice **500** to a residue having a high concentration of the desired isotope.

[0061] Referring now to FIG. 7, a top view of the target **210** (or target **110**) is shown, according to some embodiments. In the example shown, the target **210** is shown as including the first lattice **400** (the second lattice **402** may be included in some embodiments, e.g., obscured behind the first lattice **400** from the perspective of FIG. 7), a backing plate **600**, a mount **602** formed as a ring and positioned such that the first lattice **400** is between the mount **602** and the backing plate **600**, and bolts (or other coupling members, e.g., screws, clips, connectors, etc.) **604** coupling the mount **602** to the backing plate **600**.

[0062] When the bolts **604** are tightened, the mount **602** holds the first lattice **400** (or other fibrous lattice, e.g., fibrous lattice **500**) against the backing plate **600** such that the first lattice **400** is secured in position relative to the backing plate **600**. The bolts **604** can be loosened to release the first lattice **400** from the mount **602** such that the first lattice **400** can be removed for harvesting of the collected isotope from the first lattice **400**.

[0063] In some embodiments, the backing plate **600** is coupled to or part of the interior plate **212** of the magnetic rotation device **206** (actuator) of FIG. 4. In such embodiments, operation of the motor **216** to drive the exterior plate **214** causes rotation of the interior plate **212**, the backing plate **600**, the mount **602**, and the fibrous lattice (e.g., the first lattice **400**). The fibrous lattice is thereby rotatable about an axis, for example to increase a surface areas of the fibrous lattice (e.g., of the first lattice **400**) on which the ion beam is incident.

[0064] FIG. 7 shows an example in which a beam of ytterbium ions with energies of 60 kV was generated and directed at the first lattice **400**, causing the first lattice **400** to capture the ytterbium ions/atoms. FIG. 7 shows a burn area **606** within which the ytterbium was captured by the first lattice **400**. In the example shown, the ion beam causes partial burning of the first lattice **400** during ion capture. FIG. 7 also illustrates that moving the first lattice **400** while the ion beam is incident on the first lattice **400** (e.g., by rotation as described above) can allow more of the first lattice **400** to be used to capture ions, thereby increasing total collection of desired isotopes/atoms.

[0065] Referring now to FIG. 8, a top of the first lattice **400** and the second lattice **402** (partially obscured by the first lattice **400**) is shown, according to an experimental result. In the example of FIG. 8, an ion beam of ytterbium having energies of approximately 5 kV was provided incident on the first lattice **400** and the second lattice **402** (with some ions passing through the first lattice **400** to reach the second lattice **402**). A burn area **700** shows where ions were collected in the first lattice **400** and the second lattice **402**. The experiment of FIG. 8 shows that ytterbium can be captured by the first lattice **400** and the second lattice **402**. FIG. 8 also illustrates that rotating the first lattice **400** and the second lattice **402** would expose more of the fibrous lattice material to the ion beam to increase the total collection of a desired isotope by the fibrous lattice.

[0066] Referring now to FIG. 9, a top of the first lattice **400** and the second lattice **402** (partially obscured by the first lattice **400**) is shown, according to another experimental result. In the example of FIG. 9, an ion beam of ytterbium having energies of approximately 0.3 kV was provided incident on the first lattice **400** and the second lattice **402** (with some ions passing through the first lattice **400** to reach the second lattice **402**). A burn area **800** shows where ions were collected in the first lattice **400** and the second lattice **402**. Comparing the experiments of FIGS. 8-9, the beam having energies of approximately 0.3 kV provided increased capture than the beam of approximately 5 kV (as evidenced by the larger burn area **800** compared to the burn area **700**). FIGS. 8-9 thereby illustrate the advantage of reducing ion energy before collision with the target **110** as described above with reference to FIGS. 1-3. FIG. 9 also illustrates that rotating the first lattice **400** and the second lattice **402** would expose more of the fibrous lattice material to the ion beam to increase the total collection of a desired isotope by the fibrous lattice. The various features described herein thereby contribute to highly-efficient capture of desired isotope(s)/atom(s) in a

fibrous lattice material from which a residue rich in the desired particles can be easily obtained.

[0067] Although the discussion above gives a general overview of the principles of physics associated with operation of the ion production system **100**, effects of the target voltage on the ion beam, etc., it should be appreciated that the behavior of ion beams is complex and that additional or alternative theories or experimental results may be used to provide further or alternative explanation of the various advantages of the systems and methods described herein. For example, experimental results have shown that providing the target **110** with a target voltage of the same polarity as the ion beam as described above provides the advantages described herein and that the fibrous lattice materials described herein provide for efficient capture of desired isotopes such as ytterbium-176.

[0068] As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical values or idealized geometric forms provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

[0069] The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

[0070] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0071] Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

Claims

1. A method comprising: accelerating a plurality of ions toward a target comprising a target material; wherein a first portion of the plurality of ions comprise a plurality of first isotope ions; isolating the plurality of first isotope ions from a second portion of the plurality of ions; and capturing the plurality of first isotope ions with the target material, wherein the target material

comprises greater than 95% carbon.

2. The method of claim 1, wherein the plurality of ions comprise a plurality of heavy metal ions.

3. The method of claim 1, wherein: the plurality of ions comprise a plurality of ytterbium ions; and the plurality of the first isotope ions comprise a plurality of ytterbium-176 ions.

4. The method of claim 1, wherein the target material comprises carbon felt, carbon foam, carbon fiber cloth, graphite, or a combination thereof.

5. The method of claim 1, further comprising burning the target material to obtain a residue comprising the plurality of first isotope ions.

6. The method of claim 1, wherein isolating the plurality of first isotope ions comprises applying a magnetic field to the plurality of ions using a magnetic analyzer, thereby mass separating the plurality of ions; and blocking the second portion of the plurality of ions using a mass-resolving aperture positioned between the magnetic analyzer and the target, wherein the plurality of first isotope ions pass through the mass-resolving aperture.

7. The method of claim 1, wherein accelerating the plurality of ions toward the target comprises providing the plurality of ions with energies greater than 100V.

8. The method of claim 1, wherein capturing the plurality of the first isotope ions with the target material comprises decelerating the plurality of the first isotope ions by deflecting the plurality of the first isotope ions off of a plurality of fibers of the target material.

9. The method of claim 1, further comprising increasing an area of the target material that captures the plurality of first isotope ions by operating an actuator to rotate or translate the target.

10. The method of claim 9, wherein operating the actuator to rotate or translate the target further includes mechanically coupling the target to an interior plate within a vacuum chamber and magnetically coupling the interior plate to an exterior plate outside of the vacuum chamber; wherein the actuator drives rotation or translation of the exterior plate, which magnetically transfers the rotation or translation to the interior plate.

11. An ion production system comprising: an ion source configured to produce ions; a target comprising a target material; and an electrode positioned between the ion source and the target material and configured to accelerate the ions toward the target material so that the ions are incident on the target material; wherein the target material is configured to capture the ions and comprises greater than 95% carbon.

12. The ion production system of claim 11, wherein the target material comprises carbon felt, carbon foam, carbon fiber cloth, graphite, or a combination thereof.

13. The ion production system of claim 11, further comprising an actuator operable to rotate or translate the target.

14. The ion production system of claim 11, further comprising: a magnetic analyzer configured to provide a magnetic field on the ions, thereby mass separating the ions; and a mass-resolving aperture positioned between the magnetic analyzer and the target.

15. The ion production system of claim 11, wherein the ions are heavy metal ions.

16. The ion production system of claim 11, wherein the ions are ytterbium ions; and wherein at least a portion of the ytterbium ions is ytterbium-176 ions.

17. The ion production system of claim 16, further comprising: a magnetic analyzer configured to provide a magnetic field on the ytterbium ions, thereby mass separating the ytterbium ions; and a mass resolving aperture positioned between the magnetic analyzer and the target, wherein the mass resolving aperture is configured to permit the ytterbium-176 ions to pass through the mass resolving aperture such that the ytterbium-176 ions are captured with the target material.

18. The ion production system of claim 11, wherein the target is positioned within a vacuum chamber, and the system further comprises a magnetic rotation device with an interior plate positioned within the vacuum chamber and an exterior plate positioned outside of the vacuum chamber.

19. The ion production system of claim 18, wherein the target is mechanically coupled to the

interior plate; and wherein the system further comprises a motor mechanically coupled to the exterior plate.

20. The ion production system of claim 19, further comprising a cooling system that is thermally coupled to the exterior plate.
